Political Polarization and Right-wing Populism: Hypothesis Testing Exercise

Research Context:

Researchers are investigating whether exposure to polarizing political content on social media increases support for right-wing populist candidates. They randomly selected two groups of voters: one group was exposed to highly polarizing political content on a simulated social media feed for two weeks, while the control group viewed neutral political news. After the two-week period, both groups were asked about their support for a right-wing populist candidate.

Part 1: Hypotheses (5 minutes)

For this study, write down:

1. The null hypothesis (H_0): ______________________________
2. The alternative hypothesis (H_a): ______________________________

Part 2: Data Analysis (8 minutes)

Below is data collected from the experiment:

Group	Support Candidate	Don't Support	Total	Support Rate
Polarizing Content Group	42	58	100	?
Control Group	28	72	100	?

Tasks:

1. Calculate the support rate for each group.
2. Calculate the difference in support rates between the groups.
3. Make a decision using this decision rule:
   - If the difference is less than 10%, accept the null hypothesis.
   - If the difference is 10% or greater, reject the null hypothesis.

Part 3: Error Types & Sampling Error (7 minutes)

Discuss the following questions in your group:

1. In this research context, what would constitute a Type I error?
2. What would constitute a Type II error?

3. How might sampling error affect our conclusions in this study?
4. If the research team only surveyed voters from a particular age group or geographic region, what problems might arise?

Summary

Be prepared to share your group's:

- Hypotheses
- Calculation results
- Decision on whether to reject the null hypothesis
- Examples of Type I and Type II errors in this context
- How sampling error might impact the study's conclusions